

Erdős Problem #956

EP-956	L1 - Tractable	Partiellement résolu	Combinatorics
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Énoncé original (anglais, reproduction mot pour mot)

If $C, D \subseteq \mathbb{R}^2$ then the distance between C and D is defined by $\delta(C, D) = \inf_{\{c \in C \setminus d \in D\}} |c - d|$. Let $h(n)$ be the maximal number of unit distances between disjoint convex translates. That is, the maximal m such that there is a compact convex set $C \subseteq \mathbb{R}^2$ and a set X of size n such that all $(C+x)_{x \in X}$ are disjoint and there are m pairs $x_1, x_2 \in X$ such that $\delta(C+x_1, C+x_2) = 1$. Determine $h(n)$ - in particular, prove that there exists a constant $c > 0$ such that $h(n) > n^{1+c}$ for all large n .

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